# Salad AI Gateway testing and findings

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Measurements October 15 to 16 2026 UTC
Public benchmark source and results: https://github.com/James-McMenamin/salad-inference-benchmarks

## Summary

We evaluated Salad AI Gateway for MnMOracle, a game knowledge assistant that interprets informal Discord questions and answers from supplied wiki passages. Qwen3.6-35B-A3B gave the strongest Salad result in our single-request comparison: 30 of 30 grounded-answer trials passed, with a 1.93 second median complete response and an estimated $0.170 per 1,000 answer attempts.

Availability was the main obstacle in the subsequent burst tests. We observed HTTP 503 responses, timeouts, HTTP 400 responses saying the model did not exist, a separate HTTP 401 episode, and later model discovery without the requested 35B model. Successful small bursts show the endpoint can respond quickly; the failures and changing conditions prevent us from certifying a reliable concurrency or arrival-rate limit.

## Workload and method

The frozen development set contained 24 question-interpretation cases, 10 grounded-answer cases and 12 game-question classification cases. The initial comparison repeated each case three times: 138 task trials per model and 165 API calls per Salad model. These are repeated regression fixtures, not 138 independent questions. Some cases resemble prompt examples.

A grounded workflow normally makes two sequential API calls: question interpretation, then answer generation. Tests used POST https://ai.salad.cloud/v1/chat/completions, temperature 0, stream false and chat_template_kwargs.enable_thinking=false. Reader output was limited to 260 tokens and answer output to 400. Fixed prompts, sources and parser snapshots were retained. Retry policies varied by experiment and are identified below.

## Single request comparison

| Model | Interpretation | Grounding | Classifier | Answer median / p95 |
|---|---|---|---|---|
| Gemma E4B local | 57/72 | 30/30 | 33/36 | 0.79 / 1.08 s |
| Qwen3.5 9B | 36/72 | 17/30 | 24/36 | 5.83 / 14.68 s |
| Qwen3.6 35B A3B | 60/72 | 30/30 | 36/36 | 1.93 / 3.53 s |

Source A. Latency includes interpretation plus answer generation. Salad timings include network and provider overhead; the historical same-day Gemma baseline used a local RTX 3090 Ti and shared host load. This is not a controlled hardware comparison. The existing report records 18 passing harness checks and post-run frozen-input hash verification.

# Concurrency and availability

The main 35B concurrency experiment attempted 384 workflows: 96 at each of concurrency 2, 4, 8 and 16. Each level repeated a 32-question schedule in three rounds with rotated level order. No retries were enabled. All scheduled workflows were attempted, but most did not produce complete answers. Source B.

| Concurrent workflows | Completed / attempted | Rubric passes / completed | Completed median / p95 |
|---|---|---|---|
| 2 | 40/96 | 38/40 | 1.86 / 3.72 s |
| 4 | 64/96 | 64/64 | 1.56 / 3.55 s |
| 8 | 32/96 | 31/32 | 1.97 / 5.06 s |
| 16 | 1/96 | 1/1 | 3.55 / 3.55 s |

Overall, 137 of 384 workflows completed (35.7%); 134 of those completed workflows passed the automatic rubric. The latency table includes completed workflows only. The 16-concurrency timing is a single observation and is not representative.

| Concurrent workflows | HTTP 400 | HTTP 503 | Timeouts |
|---|---|---|---|
| 2 | 32 | 24 | 0 |
| 4 | 32 | 0 | 0 |
| 8 | 56 | 8 | 0 |
| 16 | 32 | 44 | 19 |

These are terminal workflow errors, not a count of all HTTP attempts. Across the run there were 152 HTTP 400 errors, 76 HTTP 503 errors and 19 timeouts. The observed HTTP 400 body reported that the model did not exist. Errors later affected lower concurrency levels as well. The endpoint recovered late in round three; concurrency alone is not a proven cause.

## Retry experiment remained incomplete

The follow-up planned 96 workflows at concurrency 16 with up to three retries per failed call, 2/4/8 second backoff plus jitter, and Retry-After handling. It stopped after 48 workflows when an HTTP 401 was observed. Of those 48, 31 completed and passed; 16 ended in HTTP 503 and one in HTTP 401. There were 99 failed API attempts and 46 workflows were retried. Completed-workflow median latency was 95.02 seconds and p95 was 102.77 seconds. This partial run is not a passing 96-workflow test. Source C.

## Recovery diagnostic

Four sequential diagnostic calls at 23:16:59 UTC on October 15 returned HTTP 200: a minimal prompt took 8.04 seconds initially and 0.51 seconds on repetition; grounded-answer calls took 0.64 and 0.73 seconds. This demonstrates recovery at that time, but does not explain the preceding failure. The saved x-request-id header values were null; completion IDs and response fingerprints were retained. Source D.

# 35B burst and arrival rate results

A workflow means a question interpretation plus grounded answer. The following runs used one set of short bursts rather than sustained traffic. Latencies in these tables include all attempted workflows, including terminal errors and timeouts; they must not be compared directly with the completed-only timings on the previous page.

| Run and workload | Passes / attempts | Errors | Median / p95 seconds |
|---|---|---|---|
| E 4 simultaneous | 4/4 | 0 | 2.39 / 4.46 |
| F 8 simultaneous | 7/8 | 0 | 2.45 / 5.09 |
| G 12 simultaneous | 8/12 | 4 | 5.53 / 91.39 |
| H 16 simultaneous | 1/16 | 15 | 90.01 / 95.59 |
| I 16 staggered | 1/16 | 15 | 90.01 / 98.01 |

All five runs above disabled retries. The staggered run started 16 workflows at 62.5 ms intervals over a nominal 937.5 ms. It still produced 15 terminal errors. The four- and eight-workflow bursts had no request errors, although one answer in the eight-workflow burst failed the rubric. Single small bursts do not establish repeatable capacity.

## Arrivals over three seconds

| Run and policy | Passes / attempts | Errors | Median / p95 seconds |
|---|---|---|---|
| J 4/sec, no retries | 10/12 | 2 | 4.31 / 91.46 |
| K 4/sec, backoff retries | 10/12 | 1 | 97.47 / 99.01 |
| L 4/sec, immediate retries | 12/12 | 0 | 1.90 / 5.63 |
| M 8/sec, no retries | 0/24 | 24 | 8.69 / 98.59 |
| N 8/sec, immediate retries | 5/24 | 19 | 8.62 / 93.17 |

Each run started four or eight new workflows at t=0, 1 and 2 seconds, giving 12 or 24 workflows total. These arrival rates are workflow starts, not API calls per second or completed answers per second. Earlier workflows could still be active when the next group started.

Run L completed 12/12 with zero failed API attempts and zero actual retries. It therefore provides no evidence that immediate retries improved reliability. In run N all five completed answers required no retries; the 19 retried workflows exhausted retries without recovery. In run K, ten workflows recovered after a 90-second timeout; all 11 completed answers were assessed as source-supported, while one recipe answer failed an exact-match singular/plural rubric check.

A prior 4/sec attempt at 23:59:58 UTC on October 15 ended with HTTP 401 on all 12 workflows and was marked failed. A later 8/sec experiment configured with a 10-second timeout stopped at model discovery before testing any workflows because the requested model was absent. Neither is silently counted as a completed successful run. Sources O and P.

# Other models and answer quality

We also tested Qwen3.5-9B and Qwen3.6-27B at nominal arrivals of 1, 2, 4 and 8 workflows per second. Each rate used three groups starting at t=0, 1 and 2 seconds, for 3, 6, 12 and 24 workflows respectively. Each model therefore attempted 45 workflows. Calls used a 10-second timeout and up to three immediate retries. Sources Q and R.

| Model | Starts/sec | Passes / attempts | Errors | Median / p95 seconds |
|---|---|---|---|---|
| 9B | 1 | 3/3 | 0 | 7.86 / 8.67 |
| 9B | 2 | 5/6 | 0 | 2.86 / 12.00 |
| 9B | 4 | 6/12 | 5 | 22.15 / 29.69 |
| 9B | 8 | 1/24 | 22 | 11.51 / 25.29 |
| 27B | 1 | 3/3 | 0 | 6.34 / 16.18 |
| 27B | 2 | 6/6 | 0 | 9.28 / 17.69 |
| 27B | 4 | 2/12 | 10 | 24.53 / 30.05 |
| 27B | 8 | 2/24 | 22 | 23.01 / 35.29 |

These timings include errors and retries. At 4 and 8 starts per second both models experienced frequent terminal errors. The low-rate cells contain only three or six workflows; they do not establish sustained reliability. Different rates cover different mixes of the ten cases, so accuracy cannot be compared as though case coverage were equal.

## 35B answer quality

In the initial comparison, the recorded assistant review checked 13 distinct 35B grounded-answer texts against the supplied sources and found them supported. Interpretation still failed on ambiguous “ss”, map follow-up context, and a misspelled NPC title. In the larger concurrency run there were three incorrect NPC abstentions. Review also found a recipe answer claiming required Enchanting skill 25 when the source distinguished Start 5 from Trivial 25; the automatic rubric did not catch that extra claim.

## 9B and 27B answer quality

In the initial 9B comparison, 12 of the 13 failed grounded-answer trials produced repetitive non-English output; the remaining failure lost a pronoun referent and declined. The later 9B rate sweep also showed misinterpretation of “wat” and a lost NPC follow-up referent, plus a singular/plural ingredient naming variation that failed the exact-match rubric. These observations describe the tested endpoint and configuration; they do not establish an underlying model defect or its cause.

The recorded assistant review of the 27B rate sweep found the nine distinct completed answer texts supported by the fixed sources. Only 13 of 45 workflows completed, so this is limited quality evidence from the answers that survived the availability failures. These were assistant reviews, not independent human adjudication.

# Findings and questions for Salad

## What the evidence supports

Qwen3.6-35B-A3B was the strongest Salad candidate in this frozen single-request comparison. Its measured answer latency and token cost were attractive for our short grounded-answer workload. Burst reliability varied sharply over time, and isolated successful runs coexisted with service-unavailable responses, long timeouts and model availability errors. We have not established a dependable maximum concurrency or sustained throughput.

## What we would like Salad to investigate

• Correlate the UTC run windows and retained completion IDs with gateway and inference logs, particularly 22:50 UTC October 15 through 00:39 UTC October 16.

• Explain why an advertised model returned HTTP 400 “model does not exist” during the main test and why the requested 35B model was later absent from model discovery. Confirm whether these were routing, deployment, availability or other events.

• Investigate the HTTP 503 and timeout periods, and clarify expected concurrency, rate limits, queuing, timeout behavior and retry guidance for this two-call workflow.

• Investigate the HTTP 401 episodes separately. The client stop message groups authorization, credit and model failures together; the observed HTTP status alone does not identify the actual cause.

• Clarify the tested 9B endpoint behavior with thinking disabled and our preserved prompts, including repetitive non-English generations.

• Confirm historical token pricing and billing treatment for timed-out or failed calls, and whether additional request identifiers can be exposed for trace correlation.

## Cost findings

Using the pricing reference captured by the benchmark, 35B cost $0.06 per million input tokens and $0.95 per million output tokens. The initial answer workload cost about $0.170 per 1,000 attempts. Completed answers at concurrency 2/4/8 cost about $0.171–$0.173 per 1,000, but that completed-answer figure excludes costs from partial failed workflows. The complete 384-workflow run recorded $0.02623435 in returned usage charges, including successful calls inside failed workflows.

The initial 9B answer workload cost $0.193 per 1,000 attempts or $0.340 per 1,000 rubric passes, using $0.07/$0.23 per million input/output tokens. The benchmark noted a discrepancy between the marketing homepage and pricing reference. These are historical usage-derived estimates, not invoice reconciliation; failed calls without usage have unknown charges. No local GPU electricity or depreciation comparison was measured. Pricing reference recorded at test time: https://docs.salad.com/ai-gateway/reference/pricing.

## Scope limits and proposed follow up

The tests used fixed wiki passages and excluded production retrieval, Discord delivery, live memory and other support calls. No streaming time-to-first-token or provider VRAM measurement was taken. Hardware and quantization at the API were undisclosed. Shared local activity and changing provider conditions remain confounders. These short development-set runs do not certify production capacity or general accuracy.

After Salad reviews the incident windows, the next useful experiment would be an agreed sustained-arrival test with documented rate limits, retry rules, server correlation IDs and success-only latency reported alongside total errors and all-attempt latency. This is a proposed follow-up; no new tests were run to prepare this report.

# Evidence index

All run paths below are relative to results/ in the public benchmark repository. Timestamps in run names are UTC. Source IDs match the report tables and narrative. Sanitized raw evidence, frozen fixtures, historical harness scripts and a portable API runner are published at https://github.com/James-McMenamin/salad-inference-benchmarks. Local machine paths are normalized; the export manifest records original and public-file hashes.

A salad-api-2026-10-15T22-25-16-854Z

B salad-api-concurrency-2026-10-15T22-50-02-529Z

C salad-api-concurrency-retry-2026-10-15T23-02-10-591Z

D salad-gateway-diagnostic-2026-10-15T23-16-59-956Z

E salad-api-four-burst-2026-10-15T23-18-29-596Z

F salad-api-eight-burst-2026-10-15T23-19-50-990Z

G salad-api-twelve-burst-2026-10-15T23-42-51-187Z

H salad-api-sixteen-burst-2026-10-15T23-20-29-112Z

I salad-api-sixteen-staggered-2026-10-15T23-38-38-515Z

J salad-api-four-per-second-2026-10-16T00-11-04-349Z

K salad-api-four-per-second-retry-2026-10-16T00-19-43-586Z

L salad-api-four-per-second-immediate-retry-2026-10-16T00-32-22-273Z

M salad-api-eight-per-second-2026-10-15T23-49-26-097Z

N salad-api-eight-per-second-immediate-retry-2026-10-16T00-33-18-303Z

O salad-api-four-per-second-2026-10-15T23-59-58-826Z

P salad-api-eight-per-second-immediate-retry-10s-2026-10-16T00-38-30-986Z

Q salad-api-rate-sweep-qwen3.5-9b-2026-10-16T00-46-03-119Z

R salad-api-rate-sweep-qwen3.6-27b-2026-10-16T00-47-35-257Z

## Files for technical review

Completed runs generally retain spec.json, receipt.json, model discovery snapshots, frozen input copies, per-level responses.jsonl and waves.jsonl, summary.json and comparison.json. A and B additionally retain review-data.json and assistant review records. C has a partial c16/summary.json and results-and-answers.txt. D has results.json. O has a failed receipt and c12/summary.json; P has a failed receipt and discovery snapshot with no workload results. Several generated comparison reports retain generic template wording; this report uses the actual schedules and distinguishes arrival rate from total workflow count.

Preparatory records are also retained in salad-api-probe-20261015T222223Z, the empty startup directory salad-api-2026-10-15T22-24-06-345Z, and salad-api-2026-10-15T22-24-27-692Z. The last failed before any completed trial with HTTP 400 “invalid request body”. They are excluded from benchmark scores. No root cause for that request rejection is established here.

Local comparison details are published in docs/local-gemma-concurrency.md. README.md and docs/METHODOLOGY.md describe reproduction and limitations. Original specifications retain source hashes; export-manifest.json identifies normalized public copies. Authentication headers, API credentials, model weights and production conversations are excluded.